COURSE 4 · CHEMISTRY

Chemistry

Printable workbook with worked solutions

64 problems across 8 modules, plus the course capstone.
Full worked solutions at the back.

Name _____ Started _____

Before you start

How to use this workbook

There are 64 problems, eight per module, followed by the course capstone. Dots show difficulty: one dot is a warm-up, two is the standard you are aiming for, three stretches towards first-year university.

Work in pencil and show your steps. Try each problem before looking at the solutions, which start on the page headed *Worked solutions*. The same problems are in the interactive workbook in the Second Pass app, which checks answers instantly. The app also has a quiz for every module and course, and spaced-repetition flashcards; a printable set of the flashcards comes with each course.

> **A good routine for each module**
>
> Problems 1–2 straight after listening to the reader. Problems 3–6 after the explorer. Problems 7–8 at the end of the week, then take the module quiz in the app and redo anything you got wrong without looking. Review the module's flashcards for a few minutes each day after that.

Module 4.1

Matter and atomic structure

4.1.1 ●○○

How many neutrons are in an atom of iron-56 (Z = 26)?

Answer: _____

4.1.2 ●○○

What defines which element an atom is?

A. number of neutrons

B. number of protons

C. mass number

D. number of shells

4.1.3 ●●○

How many electrons does a Mg^{2+} ion have? (Mg: Z = 12)

Answer: _____

4.1.4 ●●○

Give the shell arrangement of chlorine (Z = 17).

A. 2, 8, 7

B. 2, 7, 8

C. 8, 8, 1

D. 2, 8, 8

4.1.5 ●●○

Boron is 20% boron-10 and 80% boron-11. What is its relative atomic mass?

Answer: _____

4.1.6 ●●○

Carbon-12 and carbon-14 differ in

A. protons

B. electrons

C. neutrons

D. chemical behaviour

4.1.7 ●●○

How many valence electrons does sulfur (Z = 16) have?

Answer: _____

4.1.8 ●●●

Explain why isotopes of the same element have almost identical chemistry.

Module 4.2

The periodic table

4.2.1 ●○○

Elements in the same group have the same number of

A. shells

B. valence electrons

C. neutrons

D. protons

4.2.2 ●○○

Which group contains the noble gases?

A. 1

B. 2

C. 17

D. 18

4.2.3 ●●○

Which atom is larger?

A. sodium

B. chlorine

C. they are equal

D. it depends on temperature

4.2.4 ●●○

Which is the most reactive alkali metal of these?

A. lithium

B. sodium

C. potassium

D. they are equal

4.2.5 ●●○

Which is the most reactive halogen of these?

A. fluorine

B. chlorine

C. bromine

D. iodine

4.2.6 ●●○

Calcium is in group 2. What charge does its ion usually carry? (enter as a number, e.g. 2 for 2+)

Answer: _____

4.2.7 ●●○

Why does ionisation energy rise across period 3?

A. more shells

B. more protons pulling the same shell

C. fewer electrons

D. larger atoms

4.2.8 ●●●

Using electron arrangements, explain why sodium and chlorine react so readily with each other.

Module 4.3

Bonding and molecular shape

4.3.1 ●○○

What type of bonding is in magnesium oxide, MgO?

A. covalent

B. ionic

C. metallic

D. hydrogen

4.3.2 ●○○

Why do metals conduct electricity?

A. they contain ions only

B. they have a sea of free electrons

C. they are polar

D. they have hydrogen bonds

4.3.3 ●●○

What is the shape of methane, CH_4?

A. linear

B. trigonal planar

C. tetrahedral

D. bent

4.3.4 ●●○

CO_2 has polar bonds. Why is the molecule non-polar?

A. it is linear, so the dipoles cancel

B. carbon is not electronegative

C. it has lone pairs

D. it is ionic

4.3.5 ●●○

How many electrons are shared in a double bond?

Answer: _____

4.3.6 ●●○

Which molecule can form hydrogen bonds with itself?

A. CH_4

B. H_2

C. NH_3

D. CO_2

4.3.7 ●●○

Sodium chloride conducts electricity

A. as a solid

B. when molten or dissolved

C. never

D. only as a gas

4.3.8 ●●●

Explain why ice floats on water.

Module 4.4

The mole and stoichiometry

4.4.1 ●○○

How many moles are in 44 g of carbon dioxide (M = 44 g/mol)?

Answer: _____ mol

4.4.2 ●○○

What is the molar mass of sulfuric acid, H_2SO_4? (H = 1, S = 32, O = 16)

Answer: _____ g/mol

4.4.3 ●●○

How many molecules are in 9.0 g of water (M = 18 g/mol)?

Answer: _____

4.4.4 ●●○

Balance: __ H_2 + O_2 → __ H_2O. What coefficient goes in front of H_2?

Answer: _____

4.4.5 ●●○

How many grams of oxygen are needed to burn 8.0 g of methane completely? (CH_4 = 16, O_2 = 32)

Answer: _____ g

4.4.6 ●●○

What is the concentration of 0.25 mol of NaCl dissolved to make 500 mL of solution?

Answer: _____ mol/L

4.4.7 ●●○

You need 100 mL of 0.20 M solution from a 1.0 M stock. What volume of stock do you use?

Answer: _____ mL

4.4.8 ●●●

A reaction should give 50 g of product but gives 42 g. What is the percentage yield?

Answer: _____ %

Module 4.5

States of matter and gas laws

4.5.1 ●○○

A gas at 100 kPa occupies 6.0 L. It is compressed to 2.0 L at constant temperature. What is the new pressure?

Answer: _____ kPa

4.5.2 ●○○

A balloon of 2.0 L at 300 K is heated to 450 K at constant pressure. What is its new volume?

Answer: _____ L

4.5.3 ●●○

What volume does 2.0 mol of gas occupy at 300 K and 100 000 Pa? Give your answer in litres.

Answer: _____ L

4.5.4 ●●○

How many moles are in a 10 L cylinder at 500 kPa and 290 K?

Answer: _____ mol

4.5.5 ●●○

Why must temperatures be in kelvin for gas laws?

A. Celsius is less precise

B. the laws are proportional relationships measured from absolute zero

C. kelvin is larger

D. by convention only

4.5.6 ●●○

According to kinetic theory, heating a gas in a rigid container raises its pressure because

A. particles get larger

B. particles move faster and hit the walls harder and more often

C. more particles form

D. the container shrinks

4.5.7 ●●○

A tyre at 15 °C has a pressure of 240 kPa. After driving it is at 45 °C. Assuming fixed volume, what is the pressure?

Answer: _____ kPa

4.5.8 ●●●

Why do real gases deviate from the ideal gas law at high pressure and low temperature?

Module 4.6

Energy, rates and equilibrium

4.6.1 ●○○

An exothermic reaction

A. absorbs heat

B. releases heat

C. has $\Delta H > 0$

D. needs a catalyst

4.6.2 ●○○

A catalyst works by

A. raising the temperature

B. lowering the activation energy

C. being used up

D. changing ΔH

4.6.3 ●●○

Breaking bonds in a reaction needs 1370 kJ; making the new bonds releases 1652 kJ. What is ΔH?

Answer: _____ kJ

4.6.4 ●●○

Powdered marble reacts faster with acid than marble chips because

A. it is a different substance

B. it has a larger surface area

C. it is hotter

D. it is a catalyst

4.6.5 ●●○

For $N_2 + 3H_2 \rightleftharpoons 2NH_3$ (exothermic), raising the temperature shifts the equilibrium

A. right, more ammonia

B. left, less ammonia

C. not at all

D. right then left

4.6.6 ●●○

For the same reaction, increasing pressure shifts it

A. left

B. right

C. not at all

D. randomly

4.6.7 ●●○

For $A \rightleftharpoons B$, at equilibrium $[A] = 0.20$ M and $[B] = 0.80$ M. What is K?

Answer: _____

4.6.8 ●●●

Explain what 'dynamic' means in dynamic equilibrium.

Module 4.7

Acids, bases and redox

4.7.1 ●○○

What is the pH of a solution with $[H^+] = 10^{-4}$ M?

Answer: _____

4.7.2 ●○○

What is $[H^+]$ in mol/L for a solution of pH 3? Give a decimal.

Answer: _____ mol/L

4.7.3 ●●○

In $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$, which species is oxidised?

A. Zn

B. Cu^{2+}

C. Zn^{2+}

D. Cu

4.7.4 ●●○

25.0 mL of HCl is exactly neutralised by 20.0 mL of 0.100 M NaOH. What is the concentration of the HCl?

Answer: _____ mol/L

4.7.5 ●●○

A weak acid differs from a strong acid in that it

A. is always dilute

B. ionises only partially

C. has no protons

D. is a base

4.7.6 ●●○

What is the pH of 0.01 M NaOH at 25 °C?

Answer: _____

4.7.7 ●●○

What are the products of an acid reacting with a metal hydroxide?

A. salt and hydrogen

B. salt and water

C. oxide and water

D. acid and base

4.7.8 ●●●

Explain why a battery 'goes flat'.

Module 4.8

Organic chemistry essentials

4.8.1 ●○○

What is the formula of propane (3 carbons)?

A. C_3H_6

B. C_3H_8

C. C_3H_{10}

D. C_2H_6

4.8.2 ●○○

Octane is an alkane with 8 carbon atoms. How many hydrogen atoms does it have?

Answer: _____

4.8.3 ●●○

Which functional group identifies an alcohol?

A. –COOH

B. –OH

C. $–NH_2$

D. C=C

4.8.4 ●●○

Vinegar's ethanoic acid belongs to which family?

A. alcohols

B. esters

C. carboxylic acids

D. amines

4.8.5 ●●○

Butane and methylpropane are

A. the same compound

B. isomers

C. polymers

D. isotopes

4.8.6 ●●○

Proteins are polymers of

A. glucose

B. nucleotides

C. amino acids

D. fatty acids

4.8.7 ●●○

What is the molar mass of ethanol, C_2H_5OH? (C = 12, H = 1, O = 16)

Answer: _____ g/mol

4.8.8 ●●●

Why can carbon form far more compounds than any other element?

Capstone

Investigating an antacid tablet

An antacid tablet contains calcium carbonate, $CaCO_3$ ($M = 100$ g/mol), which neutralises stomach acid: $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$. A 1.00 g tablet is dropped into excess hydrochloric acid in the lab. Use moles, gas laws, pH and equilibrium thinking to analyse it.

Part 1 ●●○

The tablet is 75% calcium carbonate by mass. How many moles of $CaCO_3$ does it contain?

Answer: _____ mol

Part 2 ●●○

How many moles of HCl can it neutralise?

Answer: _____ mol

Part 3 ●●○

What volume of CO_2 is released at 298 K and 101 300 Pa? Give mL.

Answer: _____ mL

Part 4 ●●○

Stomach acid is about 0.10 M HCl. What volume of it can one tablet neutralise, in mL?

Answer: _____ mL

Part 5 ●●○

What is the pH of 0.10 M HCl?

Answer: _____

Part 6 ●●●

Why would a powdered tablet relieve indigestion faster than a whole one, even though it neutralises the same total amount of acid?

Answers

Worked solutions

Module 4.1 · Matter and atomic structure

4.1.1 ANSWER 30

1 N = A − Z = 56 − 26.

4.1.2 ANSWER B. number of protons

1 The atomic number, Z, the proton count.

4.1.3 ANSWER 10

1 Neutral Mg has 12 electrons.

2 2+ means two lost: 10.

4.1.4 ANSWER A. 2, 8, 7

1 Fill 2, then 8, then the remaining 7.

4.1.5 ANSWER 10.8

1 0.2 × 10 + 0.8 × 11.

2 = 10.8.

4.1.6 ANSWER C. neutrons

1 Isotopes differ only in neutron number.

4.1.7 ANSWER 6

1 Arrangement 2, 8, 6.

2 Outer shell: 6.

4.1.8 ANSWER Chemistry is decided by electrons, especially the valence electrons. Isotopes have the same number of protons and therefore the same number and arrangement of electrons. Only the neutron count differs, which changes the mass slightly but not how the atom bonds.

1 Which particles take part in reactions?

Module 4.2 · The periodic table

4.2.1 ANSWER B. valence electrons

1 That is why they react similarly.

4.2.2 ANSWER D. 18

1 Group 18, full outer shells.

4.2.3 ANSWER A. sodium

1 Same period; Cl has more protons pulling the same shell in.

4.2.4 ANSWER C. potassium

1 Reactivity increases down group 1.

4.2.5 ANSWER A. fluorine

1 Reactivity decreases down group 17.

4.2.6 ANSWER 2

1 Two valence electrons, lost to give Ca^{2+}.

4.2.7 ANSWER B. more protons pulling the same shell

1 Nuclear charge increases while shielding stays similar.

4.2.8 ANSWER Sodium (2,8,1) has one valence electron that is easy to remove, leaving a full shell. Chlorine (2,8,7) needs one electron to fill its outer shell. Transferring an electron gives Na^+ and Cl^-, both with full shells, and the oppositely charged ions attract strongly, releasing energy.

1 What does each atom need to reach a full shell?

Module 4.3 · Bonding and molecular shape

4.3.1 ANSWER B. ionic

1 Metal plus non-metal: electron transfer.

4.3.2 ANSWER B. they have a sea of free electrons

1 Delocalised electrons can move.

4.3.3 ANSWER C. tetrahedral

1 Four bonding pairs, no lone pairs.

4.3.4 ANSWER A. it is linear, so the dipoles cancel

1 Equal and opposite bond dipoles cancel.

4.3.5 ANSWER 4

1 Each bond is a pair; two bonds = 4 electrons.

4.3.6 ANSWER C. NH_3

1 H attached to N, with a lone pair on N.

4.3.7 ANSWER B. when molten or dissolved

1 Ions must be free to move.

4.3.8 ANSWER In ice, hydrogen bonds lock water molecules into an open hexagonal lattice with lots of space. In liquid water the hydrogen bonds keep breaking and reforming, so molecules pack closer together. Ice is therefore less dense than liquid water and floats.

1 Compare how molecules are arranged in solid and liquid.

Module 4.4 · The mole and stoichiometry

4.4.1 ANSWER 1 mol

1 n = m/M = 44/44.

4.4.2 ANSWER 98 g/mol

1 2(1) + 32 + 4(16) = 98.

4.4.3 ANSWER 3.01×10^{23}

1 n = 9/18 = 0.5 mol.

2 $N = 0.5 \times 6.02 \times 10^{23} = 3.0 \times 10^{23}$.

4.4.4 ANSWER 2

1 $2H_2 + O_2 \rightarrow 2H_2O$ balances 4 H and 2 O.

4.4.5 ANSWER 32 g

1 8 g CH_4 = 0.5 mol.

2 Needs 2 × 0.5 = 1.0 mol O_2 = 32 g.

4.4.6 ANSWER 0.5 mol/L

1 c = n/V = 0.25/0.500.

4.4.7 ANSWER 20 mL

1 $c_1V_1 = c_2V_2$: $1.0 \times V_1 = 0.20 \times 100$.

4.4.8 ANSWER 84 %

1 42/50 × 100.

Module 4.5 · States of matter and gas laws

4.5.1 ANSWER 300 kPa

1 $P_1V_1 = P_2V_2$.

2 $100 \times 6 = P_2 \times 2$.

4.5.2 ANSWER 3 L

1 $V_2 = V_1\,T_2/T_1 = 2 \times 450/300$.

4.5.3 ANSWER 49.88 L

1 $V = nRT/P = 2 \times 8.314 \times 300 \div 100\,000\ m^3$.

2 $= 0.0499\ m^3 \approx 49.9$ L.

4.5.4 ANSWER 2.074 mol

1 $n = PV/RT = 500\,000 \times 0.010 \div (8.314 \times 290)$.

2 ≈ 2.07 mol.

4.5.5 ANSWER B. the laws are proportional relationships measured from absolute zero

1 Doubling kelvin doubles volume; doubling °C does not.

4.5.6 ANSWER B. particles move faster and hit the walls harder and more often

1 Faster particles, more forceful and frequent collisions.

4.5.7 ANSWER 265 kPa

1 P/T constant; use kelvin: 288 K and 318 K.

2 240 × 318/288 ≈ 265 kPa.

4.5.8 ANSWER The ideal law assumes particles have no volume and no attractions. At high pressure, particles are packed close enough that their own volume is a significant fraction of the space. At low temperature, they move slowly enough that attractions between them matter. Both effects make real gases depart from PV = nRT, and eventually they condense.

1 Which two assumptions of the ideal model fail?

Module 4.6 · Energy, rates and equilibrium

4.6.1 ANSWER B. releases heat

1 Exo = out.

4.6.2 ANSWER B. lowering the activation energy

1 It provides an easier pathway.

4.6.3 ANSWER -282 kJ

1 ΔH = broken − made = 1370 − 1652.

4.6.4 ANSWER B. it has a larger surface area

1 More exposed particles, more collisions.

4.6.5 ANSWER B. left, less ammonia

1 Shifts in the endothermic direction, which is the reverse.

4.6.6 ANSWER B. right

1 4 gas molecules on the left, 2 on the right.

4.6.7 ANSWER 4

1 K = [B]/[A] = 0.80/0.20.

4.6.8 ANSWER At equilibrium the forward and reverse reactions are both still happening, at equal rates. Molecules are constantly converting in both directions, so on a molecular level the system is busy, but the overall concentrations stay constant.

1 Are the reactions stopped or balanced?

Module 4.7 · Acids, bases and redox

4.7.1 ANSWER 4

1 pH = $-\log(10^{-4}) = 4$.

4.7.2 ANSWER 0.001 mol/L

1 $[H^+] = 10^{-3}$.

4.7.3 ANSWER A. Zn

1 Zn loses two electrons.

4.7.4 ANSWER 0.08 mol/L

1 Moles NaOH = 0.100 × 0.0200 = 0.00200.

2 1:1 ratio; c = 0.00200/0.0250 = 0.080 M.

4.7.5 ANSWER B. ionises only partially

1 Strength is about extent of ionisation.

4.7.6 ANSWER 12

1 $[OH^-] = 10^{-2}$; $[H^+] = 10^{-14}/10^{-2} = 10^{-12}$.

2 pH = 12.

4.7.7 ANSWER B. salt and water

1 Neutralisation.

4.7.8 ANSWER A battery is a redox reaction separated into two halves. As it runs, the reactants are used up: the substance being oxidised and the one being reduced are converted to products. Once they are depleted, the electron 'push' between the two halves drops and the current stops. Recharging drives the reaction backwards with an external supply.

1 What is being used up?

Module 4.8 · Organic chemistry essentials

4.8.1 ANSWER B. C_3H_8

1 C_nH_{2n+2} with n = 3.

4.8.2 ANSWER 18

1 2(8) + 2 = 18.

4.8.3 ANSWER B. –OH

1 Hydroxyl group.

4.8.4 ANSWER C. carboxylic acids

1 –COOH group.

4.8.5 ANSWER B. isomers

1 Same formula, C_4H_{10}; different structures.

4.8.6 ANSWER C. amino acids

1 Amino acids joined by peptide bonds.

4.8.7 ANSWER 46 g/mol

1 2(12) + 6(1) + 16 = 46.

4.8.8 ANSWER Carbon forms four strong covalent bonds, so each atom can connect in several directions. Carbon–carbon bonds are strong enough to build stable long chains, branches and rings. It also bonds well with H, O, N and other elements, and can form single, double and triple bonds. Together these allow an almost limitless variety of structures.

1 Think about bond number, bond strength and chains.

Capstone · Investigating an antacid tablet

Part 1 ANSWER 0.0075 mol

1 0.75 g ÷ 100 g/mol.

Part 2 ANSWER 0.015 mol

1 Ratio 1 $CaCO_3$: 2 HCl.

2 $2 \times 0.0075 = 0.015$ mol.

Part 3 ANSWER 183.4 mL

1 $n(CO_2) = 0.0075$ mol.

2 $V = nRT/P = 0.0075 \times 8.314 \times 298 \div 101\,300\ m^3 \approx 1.83 \times 10^{-4}\ m^3 \approx 183$ mL.

Part 4 ANSWER 150 mL

1 $V = n/c = 0.015 \div 0.10 = 0.15$ L.

Part 5 ANSWER 1

1 Strong acid: $[H^+] = 0.10 = 10^{-1}$.

2 pH = 1.

Part 6 ANSWER Powder has a much larger surface area, so acid particles collide with calcium carbonate far more often and the reaction rate is higher. The total amount neutralised is set by the moles of $CaCO_3$, which is unchanged; only the speed differs.

1 Which factor affects rate but not amount?